Look from Above: Satellite-Guided Generative Mapping for Robust Pseudo-Labeling

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Abstract. Online vectorized HD map construction is essential for autonomous driving, yet its scalability is limited by the high cost of dense, geographically diverse map annotations. Pseudo-label collection offers an attractive alternative, but the resulting supervision is often noisy and incomplete due to occlusions and partial observability. We propose **Pseudo-Label Online Mapping**, a diffusion-based pseudo-label generator that combines two complementary priors: a BEV representation reconstructed from multi-view camera semantics and an aligned satellite-map patch that provides global layout cues. Our model fuses dual-stream BEV features and formulates map prediction as diffusion-based set denoising over polyline queries, producing vectorized map elements with confidence scores. To suppress supervision noise from unobserved regions, we restrict training and evaluation to the valid area of the reconstructed BEV. On nuScenes, our pseudo labels achieve **25.1** mean AP on the observed area (AP_{ped}=12.5, AP_{div}=31.8, AP_{bdry}=31.0), outperforming strong pseudo-label collection baselines (18.8/18.1 mean AP). We further validate the utility of our pseudo labels by supervising a MapTR-style downstream online mapping model on **[downstream setting]**.

Keywords: Pseudo Label · Satellite Map · Online Mapping

1 Introduction

High-definition (HD) maps provide critical geometric and semantic priors for autonomous driving, supporting robust localization, planning, and safety. However, maintaining HD maps at scale is expensive and slow, motivating *online vectorized mapping* that predicts local map elements on the fly from on-board sensors [8,11,14,17,24]. Compared to rasterized bird's-eye-view (BEV) segmentation, vectorized representations (e.g., polylines for dividers and boundaries) are compact, topology-friendly, and more directly usable by downstream modules, but they remain challenging to learn due to sparse supervision and ambiguity under partial observations.

A major bottleneck for scaling vectorized online mapping is annotation scarcity. While datasets such as nuScenes enable objective evaluation with ground-truth map vectors [1], collecting and maintaining such supervision for new cities and domains is impractical. This has spurred interest in generating supervision automatically via *pseudo-label collection* [16]. Yet pseudo labels derived from local observations are often noisy and incomplete due to occlusions, limited range, and reconstruction artifacts, which can introduce spurious geometry and inconsistent topology, especially outside the reliably observed region.

We propose **Pseudo-Label Online Mapping**, a diffusion-based pseudo-label generator that leverages dual complementary priors. The first input stream is a BEV representation reconstructed from multi-view camera semantics: we obtain semantic cues via Mask2Former [4] and build an aligned BEV signal following common multi-view-to-BEV paradigms [10, 19, 27], with a Gaussian-based reconstruction component [9]. The second stream is a satellite-map patch aligned to the same road segment, obtained from the nuScenes map resources [1]. Both streams are encoded with ResNet-50+FPN backbones and fused at the feature level. On top of the fused BEV features, a transformer-based vector head predicts map elements via cross-attention from learnable map queries, and a diffusion formulation iteratively denoises a set of polyline proposals into coherent vectorized pseudo labels with confidence scores [2, 3, 5, 6, 17, 18, 21].

Crucially, we focus on measuring pseudo-label quality on nuScenes [1]. We train the diffusion model with GT map vectors and then use it as a pseudo-label generator. Following the standard practice in pseudo-label collection, we evaluate on the *observed area only* by masking unobserved regions using the valid region of the reconstructed BEV, reducing noise from occlusions and surrounding clutter [16]. Under this setting, our pseudo labels achieve mean AP = 25.1 on nuScenes (AP_{ped} = 12.5, AP_{div} = 31.8, AP_{bdry} = 31.0), outperforming the PseudoMapTrainer pseudo-label collection baselines [16]: Single-trip (18.8) and Multi-trip (18.1) on the observed area.

Finally, we use these pseudo labels to supervise a MapTR-style downstream online mapping model [11] on **[Downstream dataset/setting + metric]**, demonstrating improved downstream mapping performance. *(Results will be filled once finalized.)*

2 Related Works

2.1 Online HD Map Construction

The paradigm of High-Definition (HD) map construction has undergone a significant transition from offline manual curation to real-time, onboard perception. Early dominant approaches predominantly treated mapping as a dense semantic segmentation task in the Bird's-Eye-View (BEV) space. Foundational works such as LSS [19], CVT [27], and BEVFormer [10] established the feasibility of transforming multi-view camera images into a unified top-down representation using depth estimation or transformer-based spatial cross-attention. However, these grid-based rasterized outputs, while visually intuitive, often fail to capture the high-level instance connectivity and precise lane geometry necessitated by downstream motion planners.

To bridge the gap between perception and planning, research has shifted toward vectorized representation learning, where road elements are modeled as structured geometric primitives. VectorMapNet [14] first conceptualized this by employing an auto-regressive transformer decoder to predict coordinates sequentially, effectively treating map elements as polyline sequences. While pioneering, its auto-regressive nature suffered from high inference latency and error accumulation. To overcome these limitations, MapTR [11] introduced a permutation-invariant modeling approach, treating map elements as sets of equivalent points. By utilizing a hierarchical bipartite matching loss, MapTR achieved a stable end-to-end training objective, establishing itself as a robust baseline for structured mapping. Its successor, MapTRv2 [12], further enhanced the geometric constraints by introducing auxiliary perspective-view supervision and decoupled point-line queries, pushing the boundaries of spatial accuracy.

Building on these static frameworks, recent efforts have focused on addressing the challenges of temporal stability and long-range perception. Since online mapping is inherently an incremental process, StreamMapNet [24]and BeMapNet [20] integrated temporal buffers and recurrent mechanisms to propagate historical BEV features. This temporal fusion effectively alleviates the "flickering" effect caused by transient occlusions and improves the consistency of predicted lane boundaries.

2.2 Learning from Unlabeled Data and Pseudo-labels

The scalability of online mapping is fundamentally constrained by the high cost and geographical sparsity of manual HD map annotations. To circumvent this, recent research has pivoted toward label-free and weakly-supervised learning paradigms. PseudoMapTrainer [16] represents a milestone in this direction by proposing a pipeline that eliminates the need for ground-truth HD maps during training. This approach synergizes 3D Gaussian Splatting (3DGS) [9] with pre-trained 2D semantic segmentation networks, such as Mask2Former [4], to perform high-fidelity neural reconstruction of the road surface from multi-view video sequences.

Unlike previous self-supervised methods that rely on simple geometric constraints, PseudoMapTrainer extracts structured vectorized pseudo-labels directly from the rendered semantic consistency. To address the inherent noise and spatial incompleteness in these labels—often stemming from sensor blind spots and dynamic object occlusions—the authors introduced a Mask-aware Assignment algorithm and a robust loss function. This mechanism aligns with the principles of Mask-guided Learning [13], allowing the model to focus on regions with high-confidence observations. Furthermore, as explored in MapGS [25], using such neural radiance fields or splatting techniques for data augmentation can significantly enhance a model's generalization to "in-the-wild" crowdsourced data. By enabling pre-training on vast unlabeled datasets, this paradigm fosters cross-city transferability and robust performance in novel geographical domains without human intervention.

2.3 Generative Diffusion Models for Vectorized Mapping

The construction of online HD maps is traditionally framed as a deterministic regression task. However, standard point-set regression models often struggle to capture the inherent aleatoric uncertainty and topological ambiguities—such as worn-out lane markings or severe occlusions. To address these challenges, we transition from deterministic estimation to a generative paradigm, leveraging Denoising Diffusion Probabilistic Models [6] to learn the full probability distribution of vectorized map elements. The application of diffusion in the spatial

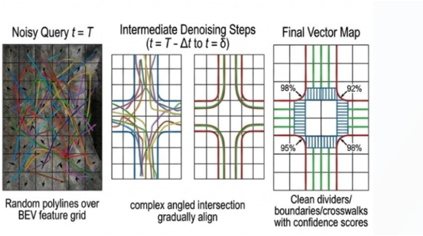

Fig. 1: diffusion

domain has evolved from occupancy and rasterized representations to structured vector outputs. Initial efforts [22] [23] focused on 3D occupancy grids or semantic raster layers conditioned on aerial imagery [8, 15, 26]. More recently, DiffMap [7] extended this to online raster map prediction from on-road views, utilizing latent diffusion to refine segmentation masks. However, rasterized outputs require post-processing and lack the structured connectivity needed for motion planning. Our work represents a significant shift toward Direct Vectorized Generation, building on the success of diffusion-based object detection, such as DiffusionDet [3], which demonstrated that bounding boxes could be effectively treated as denoised vector primitives.

PolyDiffuse [2] utilizes a Guided Set Diffusion Model to enhance the accuracy of polygonal shapes. However, PolyDiffuse primarily serves as a post-processing refinement step on top of coarse predictions from existing models like MapTR [11]. Consequently, its performance remains upper-bounded by the initial baseline, and it lacks the ability to generate diverse, independent samples from the underlying map distribution. In contrast, MapDiffusion formulates the entire mapping task as a generative process.

3 Method

We study automatic HD map labeling from vehicle-mounted camera data without relying on manually annotated HD maps. Given multi-view onboard images and an aligned satellite image patch, our goal is to generate a local vectorized HD map that is both faithful to onboard observations and structurally complete in areas that may be partially visible or ambiguous from the driving views alone. To this end, we propose a two-stage framework. In the first stage, we transform perspective-view onboard images into a pseudo-BEV map through semantic extraction, road surface reconstruction, and BEV rendering. In the second stage, we fuse the resulting pseudo-BEV representation with satellite imagery and employ a diffusion-based transformer decoder to generate the final vectorized HD map.

The key idea of our method is to decouple *observation grounding* from *map completion and vectorization*. The pseudo-BEV constructed in Stage I preserves geometry-aware evidence distilled from onboard visual observations in a top-down coordinate frame. However, due to occlusion, limited visibility, and reconstruction errors, this intermediate representation may still contain missing or unreliable structures. We therefore introduce a satellite-guided generative mapping stage that leverages external top-down priors to refine, complete, and regularize the map before decoding it into structured vector elements. This design combines the advantages of onboard observation consistency, satellite-based structural context, and generative map prediction.

3.1 Problem Formulation

Let

$$\mathcal{I} = \{I^{(1)}, \dots, I^{(M)}\}$$

denote the synchronized multi-view images captured by M onboard cameras at a target timestamp. We additionally assume access to a satellite image patch

$$S \in \mathbb{R}^{H_s \times W_s \times 3},$$

which is geographically aligned to the local area around the ego vehicle. Our objective is to automatically construct a local vectorized HD map

$$\hat{\mathcal{M}} = \{\hat{\mathcal{P}}_1, \dots, \hat{\mathcal{P}}_N\},$$

where each map element $\hat{\mathcal{P}}_n$ is represented as a polyline or polygon together with its semantic category, such as lane divider, road boundary, or pedestrian crossing.

Our framework is composed of two sequential stages. First, we build a pseudo-BEV map from onboard visual observations:

$$B_p = \Phi_{\text{pseudo}}(\mathcal{I}), \tag{1}$$

where Φ_{pseudo} denotes a pseudo-BEV construction pipeline consisting of perspective-view semantic extraction, road surface reconstruction, and orthographic BEV rendering. The resulting B_p serves as an intermediate map-space observation derived purely from onboard images.

Next, we combine the pseudo-BEV map B_p with the aligned satellite patch S and predict the final vectorized HD map:

$$\hat{\mathcal{M}} = \Phi_{\text{gen}}(B_p, S), \tag{2}$$

where Φ_{gen} is a satellite-guided generative mapping network. Unlike deterministic vector decoders that output a single map hypothesis, we formulate this stage as conditional generative prediction:

$$\hat{\mathcal{M}} \sim p_\theta(\mathcal{M} \mid B_p, S). \tag{3}$$

By integrating satellite imagery, our approach compensates for the incompleteness of pseudo-BEV features. The large-scale top-down perspective provides essential structural constraints that facilitate the recovery of missing elements and ensure geometric consistency.

3.2 Overview

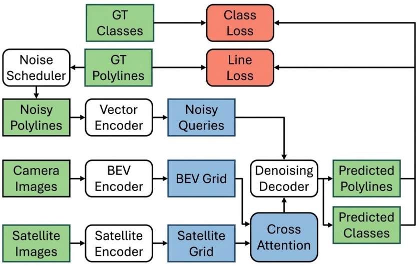

Fig. 2: Workflow

Figure 3 shows the overall pipeline of our method. Starting from multi-view onboard images, Stage I constructs a pseudo-BEV raster representation in the local ego-centric coordinate frame. Compared with raw perspective-view imagery, this intermediate representation is geometrically closer to the target HD map and is therefore more suitable for downstream map reasoning. Nevertheless, the pseudo-BEV is only an observation-driven approximation of the local map: it may suffer from incomplete coverage, occluded regions, or artifacts introduced during semantic extraction and 3D reconstruction.

To address these limitations, Stage II introduces a satellite-guided generative mapping module. Specifically, the pseudo-BEV image B_p and the aligned satellite image S are processed by two separate visual encoders:

$$F_b = E_b(B_p), \qquad F_s = E_s(S), \tag{4}$$

where E_b and E_s denote the pseudo-BEV and satellite backbones, respectively. The two feature streams are then fused by a cross-modal interaction module:

$$F_f = \mathcal{F}_{\text{fuse}}(F_b, F_s), \tag{5}$$

where $\mathcal{F}_{\text{fuse}}(\cdot)$ is implemented with cross-attention to inject complementary satellite priors into the pseudo-BEV representation.

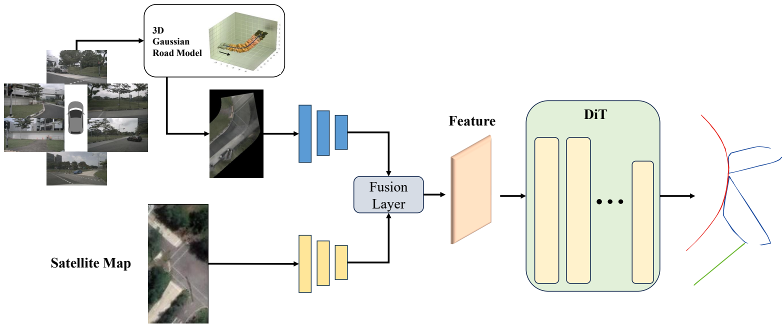

Fig. 3: Overview of the proposed pipeline.

Based on the fused feature F_f, we adopt a diffusion-based transformer decoder to generate the final vectorized HD map. Instead of directly regressing a deterministic set of map elements, the decoder starts from noisy vector queries and iteratively denoises them under the condition of the fused representation:

$$\hat{\mathcal{M}} = D_\theta(Z_T, F_f), \tag{6}$$

where Z_T denotes the noisy initial map representation at diffusion step T, and D_θ is the denoising transformer. In this way, the model directly outputs structured vectorized map elements, avoiding hand-crafted raster-to-vector post-processing in the final prediction stage.

Overall, our framework can be viewed as *pseudo-BEV construction followed by satellite-guided generative vector mapping*. Stage I extracts geometry-aware map evidence from onboard observations, while Stage II complements and regularizes this evidence with satellite priors and performs structured generative decoding in vector space. This division of labor is important: it preserves the observation consistency of the onboard sensing pipeline while enabling the final map label to benefit from stronger global context and generative modeling capacity.

3.3 Stage I: Pseudo-BEV Construction

The first stage converts multi-view onboard images into a pseudo-BEV map that serves as an intermediate, geometry-aware map observation. Starting from perspective-view camera images, we first extract road-relevant semantics, then reconstruct the observable road surface in 3D, and finally render the reconstructed semantics into the bird's-eye-view space. This stage transforms image-space observations into a map-space representation that is better aligned with the final HD map prediction target.

Perspective-view semantic extraction. Given synchronized multi-view images

$$\mathcal{I} = \{I^{(1)}, \ldots, I^{(M)}\},$$

we apply a semantic segmentation model to each camera view to identify map-relevant regions and structures in the scene. The segmentation output includes semantic categories useful for map construction. Formally, the semantic predictions are written as

$$\mathcal{S} = \{S^{(1)}, \ldots, S^{(M)}\}, \qquad S^{(m)} = \Psi_{\text{seg}}\left(I^{(m)}\right), \tag{7}$$

where Ψ_{seg} denotes the perspective-view segmentation network and $S^{(m)}$ is the semantic mask for the m-th camera. These segmented observations provide dense semantic evidence from the onboard sensing views.

3D Gaussian road reconstruction. The segmented multi-view observations are then lifted into 3D and represented with semantic Gaussians anchored to the local road surface. Using camera calibration and ego-pose information, we project the semantic evidence from image space into a shared world coordinate frame and optimize a set of 3D Gaussian primitives that describe the observable road region. Each Gaussian encodes its spatial position, geometric extent, and associated semantic attributes, allowing the reconstructed scene representation to preserve both local surface structure and category information. Let Ψ_{rec} denote this Gaussian-based reconstruction module. We express the reconstructed representation as

$$\mathcal{G} = \Psi_{\text{rec}}(\mathcal{S}), \tag{8}$$

where $\mathcal{G} = \{g_i\}_{i=1}^{N_g}$ denotes the set of reconstructed semantic Gaussian primitives. Compared with directly aggregating 2D observations on the ground plane, the Gaussian representation provides a continuous and multi-view consistent description of the observable road geometry while alleviating the perspective distortion of raw camera images.

BEV rendering via Gaussian splatting. After obtaining the reconstructed Gaussian scene representation $\mathcal{G}$, we render it into a top-down Cartesian grid centered at the ego vehicle using Gaussian splatting. Specifically, the semantic Gaussians are orthographically projected onto the BEV plane and accumulated on a regular raster grid, yielding a pseudo-BEV semantic map

$$B_p = \Psi_{\text{bev}}(\mathcal{G}), \tag{9}$$

where Ψ_{bev} denotes the BEV rendering operator based on Gaussian splatting, and

$$B_p \in \mathbb{R}^{H_b \times W_b \times C_b}$$

is the resulting BEV raster representation. Each grid cell in B_p aggregates the semantic contribution of projected Gaussians at the corresponding ground-plane location, and therefore encodes the semantic occupancy or semantic confidence in BEV space. Compared with raw perspective-view images, B_p is already represented in the same top-down coordinate system as the target HD map, making it a more suitable intermediate map-space observation for downstream vectorized map generation.

Combining the above steps, the pseudo-BEV construction process can be summarized as

$$B_p = \Psi_{\text{bev}}\big(\Psi_{\text{rec}}(\Psi_{\text{seg}}(\mathcal{I}))\big). \tag{10}$$

Although B_p provides a geometry-aware and semantics-rich map observation, it is not necessarily complete. Its quality is fundamentally bounded by the visibility of onboard cameras and the coverage of the observed scene. Occlusions, distant regions, sparse evidence, and reconstruction errors can all lead to missing or unreliable BEV structures. For this reason, we do not directly treat B_p as the final HD map label. Instead, we use it as an intermediate observation in map space, which is subsequently refined and completed in Stage II through fusion with aligned satellite imagery.

3.4 Stage II: Satellite-guided Generative Vector Mapping

Stage II refines and completes the pseudo-BEV observation B_p by incorporating aligned satellite priors and generating a vectorized HD map in a conditional generative manner. Given B_p from Stage I and the satellite patch S, we design a dual-branch fusion encoder and a diffusion-based vector decoder. Crucially, diffusion is performed in the *output vector space*: during training we perturb the ground-truth map polylines and learn to recover the *clean coordinates*.

Dual-branch encoding. We encode the pseudo-BEV raster and the satellite image with two modality-specific backbones:

$$F_b = E_b(B_p), \qquad F_s = E_s(S), \tag{11}$$

where E_b and E_s are implemented as convolutional encoders (e.g., ResNet-style backbones) followed by linear projections to a shared embedding dimension d. The pseudo-BEV branch captures observation-grounded local semantics, while the satellite branch provides complementary top-down structural context.

Cross-modal fusion by cross-attention. To inject satellite priors into the pseudo-BEV representation in a spatially adaptive manner, we employ cross-attention fusion. Let $\tilde{F}_b \in \mathbb{R}^{N_b \times d}$ and $\tilde{F}_s \in \mathbb{R}^{N_s \times d}$ be the flattened token sequences of F_b and F_s. We update pseudo-BEV tokens by attending to satellite tokens:

$$\Delta\tilde{F}_b = \text{CrossAttn}(\tilde{F}_b, \tilde{F}_s, \tilde{F}_s), \tag{12}$$

and obtain the fused representation with a residual connection:

$$\tilde{F}_f = \tilde{F}_b + \Delta\tilde{F}_b, \qquad F_f = \text{Reshape}(\tilde{F}_f). \tag{13}$$

The fused feature F_f serves as the conditioning signal for generative vector map decoding.

Vectorized map parameterization. We represent a local HD map as a set of L vector elements

$$\mathcal{M} = \{(c_\ell, \mathbf{p}_\ell)\}_{\ell=1}^{L},$$

where c_ℓ is the semantic class and

$$\mathbf{p}_\ell = \big[(x_{\ell,1}, y_{\ell,1}), \ldots, (x_{\ell,K}, y_{\ell,K})\big] \in \mathbb{R}^{K\times 2}$$

denotes the polyline/polygon control points in BEV coordinates. As in query-based vector mapping, we use a fixed number L of map queries; if the ground-truth contains fewer than L elements, the remaining slots are treated as `no-object`.

Forward diffusion in vector space. Let $X_0 = \{\mathbf{p}_1, \ldots, \mathbf{p}_L\}$ collect the ground-truth geometry (coordinates) of all map elements. We apply the forward diffusion process directly on X_0 by injecting Gaussian noise:

$$X_t \sim q(X_t \mid X_0), \qquad t = 1, \ldots, T, \tag{14}$$

where T is the number of diffusion steps. In practice, this corresponds to sampling a noise level indexed by t and perturbing the 2D control-point coordinates accordingly.

Denoising decoder and clean-coordinate prediction. We adopt a diffusion transformer (DiT-style) denoising decoder D_θ that takes the noisy vectors X_t, the diffusion step t, and the fused condition F_f as inputs, and predicts the *clean coordinates*:

$$\hat{X}_0 = D_\theta(X_t, t, F_f). \tag{15}$$

Concretely, D_θ is implemented as a transformer decoder over L map queries. The noisy vectors X_t are embedded and injected into the query stream, while F_f provides conditioning via cross-attention (or equivalently, by treating F_f tokens as memory). In addition to geometry denoising, a classification head predicts per-query semantic labels:

$$\hat{c}_\ell = H_{\text{cls}}(h_\ell), \qquad \hat{\mathbf{p}}_\ell = H_{\text{geo}}(h_\ell), \tag{16}$$

where h_ℓ denotes the final hidden state for the ℓ-th query, and H_{geo} outputs the clean control-point coordinates (i.e., the corresponding slice of $\hat{X}_0$).

Inference. At inference time, we initialize X_T with random Gaussian noise and iteratively apply the reverse denoising process conditioned on F_f:

$$X_{t-1} \leftarrow D_\theta(X_t, t, F_f), \qquad t = T, \ldots, 1, \tag{17}$$

and obtain the final vectorized HD map prediction

$$\hat{\mathcal{M}} = \{(\hat{c}_\ell, \hat{\mathbf{p}}_\ell)\}_{\ell=1}^{L}. \tag{18}$$

Training with Hungarian matching. Since vectorized map prediction is a set prediction problem, we use Hungarian matching to associate predicted queries with ground-truth elements. Let π denote a permutation that matches predictions to targets by minimizing a bipartite matching cost:

$$\pi^* = \arg\min_{\pi \in \Pi} \sum_{\ell=1}^{L} \mathcal{C}\big(\hat{c}_\ell, \hat{\mathbf{p}}_\ell;\ c_{\pi(\ell)}, \mathbf{p}_{\pi(\ell)}\big), \tag{19}$$

where Π is the set of all permutations and $\mathcal{C}(\cdot)$ combines classification and geometry terms. Following common practice in vectorized mapping decoders, we compute the training losses on the matched pairs. In particular, we supervise clean-coordinate prediction with a geometry loss $\mathcal{L}_{\text{geo}}$ (e.g., ℓ_1 over control points), and use a classification loss $\mathcal{L}_{\text{cls}}$ consistent with standard query-based vector mapping decoders. The overall objective for Stage II is

$$\mathcal{L} = \mathcal{L}_{\text{geo}}(\hat{X}_0, X_0; \pi^*) + \lambda_{\text{cls}}\, \mathcal{L}_{\text{cls}}(\hat{c}, c; \pi^*), \tag{20}$$

where λ_{cls} balances geometry and semantic supervision.

4 Experiments

4.1 Experimental Setup

Dataset and split. We conduct all experiments on nuScenes [1] and follow the geographically disjoint train/validation split adopted by PseudoMapTrainer [16] (i.e., no geospatial overlap between splits). We evaluate three map element classes: pedestrian crossings (*ped*), lane dividers (*div*), and road boundaries (*bdry*).

Metrics. We report average precision (AP) based on Chamfer distance with thresholds $\{0.5, 1.0, 1.5\}$ meters and mean AP (mAP) across the three classes, following standard practice in online vectorized mapping.

Observed-area-only protocol. Pseudo-labels collected from local observations do not reliably cover the full BEV range. Following the pseudo-label collection protocol in PseudoMapTrainer [16], we primarily evaluate pseudo-label quality on the *observed area only*. Concretely, we derive a valid-region mask from the reconstructed BEV (indicating regions with sufficient visual evidence) and ignore unobserved regions when matching predictions to ground-truth vectors, which reduces spurious geometry caused by occlusions and surrounding clutter.

Implementation details. Unless stated otherwise, we follow the training recipe commonly used in recent diffusion-based mapping work (24 epochs, batch size 1 per GPU, AdamW with cosine schedule). The BEV region of interest covers $60\,\text{m} \times 30\,\text{m}$ and is discretized into a 100×50 grid. For diffusion, we use a cosine noise schedule with $T{=}1000$ steps; during inference we use DDIM sampling

Table 1: Pseudo-label quality on nuScenes validation under the observed-area-only protocol. PseudoMapTrainer baselines are taken from [16] on the same split and metric.

Method	AP_{ped}	AP_{div}	AP_{bdry}	mAP
PseudoMapTrainer (Single-trip) [16]	23.6	6.0	26.8	18.8
PseudoMapTrainer (Multi-trip) [16]	25.8	2.3	26.2	18.1
Ours (Dual priors + diffusion)	18.5	**35.3**	**37.3**	**30.1**

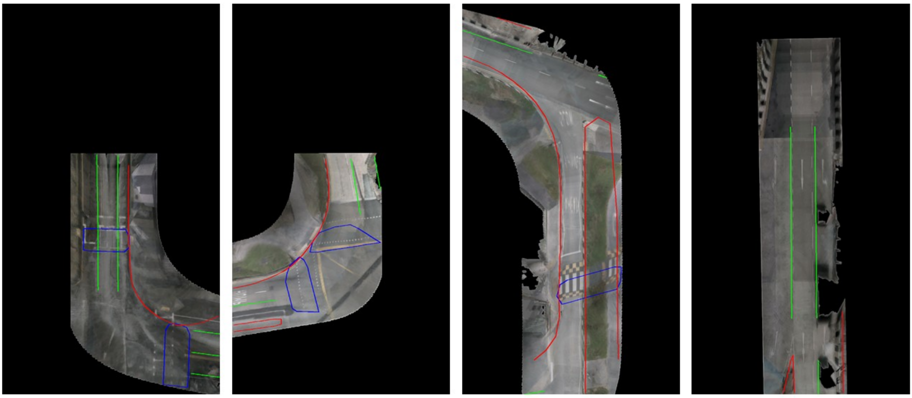

Fig. 4: Qualitative pseudo-label visualization on nuScenes (observed area only). Colors: pedestrian crossings (blue), dividers (green), boundaries (red).

with k=5 steps and η=0.5, and keep predictions with confidence above a fixed threshold. In our dual-stream design, the BEV reconstruction stream and the satellite-map stream use the *same* encoder in each setting (ResNet-50+FPN by default), ensuring a controlled comparison.

4.2 Pseudo-Label Quality on nuScenes

We train our diffusion model with ground-truth (GT) map vectors on the training split and then use the trained model as a pseudo-label generator on the validation split. Table 1 reports pseudo-label quality under the observed-area-only protocol. Our diffusion-based generator substantially improves mAP over pseudo-label collection baselines, with particularly large gains on dividers and boundaries.

4.3 Ablations

We analyze two key components: (i) the satellite-map prior, and (ii) encoder capacity. For each variant, both streams share the same encoder to keep the

Table 2: Ablations on nuScenes validation (observed area only). "Sat" indicates whether the satellite-map prior is used. "CNN" denotes a lightweight convolutional encoder; "ResNet" denotes ResNet-50+FPN.

Variant	AP_{ped}	AP_{div}	AP_{bdry}	mAP
w/o Sat + CNN	7.6	17.4	20.3	15.1
w/o Sat + ResNet	10.7	23.2	28.4	20.8
w/ Sat + CNN	12.5	25.9	29.6	22.7
w/ Sat + ResNet	**18.5**	**35.3**	**37.3**	**30.1**

Table 3: Downstream online mapping on nuScenes validation. Upper block: models trained with GT vectors. Lower block: MapTRv2 trained with pseudo labels.

Supervision / Model	AP_{ped}	AP_{div}	AP_{bdry}	mAP
GT: VectorMapNet [14]	13.7	13.5	14.9	14.0
GT: MapTR [11]	14.4	16.0	26.7	19.0
GT: MapVR	17.0	16.3	27.6	20.3
GT: StreamMapNet [24]	25.8	23.0	29.5	26.1
GT: MapTRv2 [12]	23.8	19.5	32.7	25.4
Pseudo: MapTRv2 + PseudoMapTrainer [16]	12.3	3.8	8.3	8.2
Pseudo: MapTRv2 + Ours	10.5	**16.2**	**20.1**	**15.6**

architecture symmetric. As shown in Table 2, both the satellite prior and a stronger encoder contribute consistently, and their effects are complementary.

4.4 Downstream Online Mapping with Pseudo Labels

Pseudo-label quality is only meaningful if it transfers to downstream training. We therefore train a MapTRv2-style online mapping model [12] using pseudo labels on the training split and evaluate against GT on the validation split. Table 3 shows that our pseudo labels yield a large improvement over training with PseudoMapTrainer pseudo labels, substantially narrowing the gap to GT supervision. All numbers are produced on the same split and metric, ensuring a fair comparison.

Qualitative results. We provide qualitative comparisons of pseudo labels in Fig. **[X]**. Our method produces more coherent topology and fewer spurious structures in partially observed scenes, consistent with the quantitative improvements under the observed-area-only protocol.

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